Class XII Session 2025-26
Subject - Physics
Sample Question Paper - 1

Time Allowed: 3 minutes

General Instructions:-

1. There are 15 questions in all. All questions are compulsory.
2. This question paper has five sections: Section A.
3. All the sections are compulsory.
4. **Section A** contains sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each, **Section B** contains five questions of two marks each, **Section C** contains seven questions of three marks each, **Section D** contains two case study based questions of four marks each and **Section E** contains three long answer questions of five marks each.
5. There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
6. Use of calculators is not allowed.

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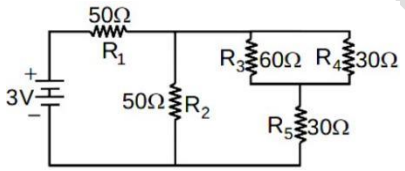
Section A

1. At which temperature, a pure semiconductor behaves slightly as a conductor? **[1]**

   a) room temp. b) both low and room temp.

   c) high temp. d) low temp.

2. In circuit shown below, the resistance are given in ohms and the battery is assumed ideal with emf equal to 3 volt. The voltage across the resistance R_4 is: **[1]**



   a) 1.2 V b) 0.6 V

   c) 0.4 V d) 1.5 V

3. Reflecting telescope utilizes **[1]**

   a) Prism b) Convex mirror

   c) Both Convex mirror and Concave mirror d) Concave mirror

4. A magnet is cut in three equal parts by cutting it perpendicular to its length. The time period of original magnet is T0 in a uniform magnetic field B. Then, the time period of each part in the same magnetic field is **[1]**

a) None of these

b) $\frac{T_0}{2}$

c) $\frac{T_0}{3}$

d) $\frac{T_0}{4}$

5. Capacitance of a conductor is $1\mu F$. What charge is required to raise its potential to 100 V? **[1]**

a) $200\mu C$

b) $50\mu C$

c) $100\mu C$

d) $150\mu C$

6. The magnetic force on a point charge is $\vec{F} = q(\vec{v} \times \vec{B})$, The dimensions of $\vec{B}$ are: (Here q = electric charge, $\vec{v}$ = velocity of point charge, $\vec{B}$ = magnetic field) **[1]**

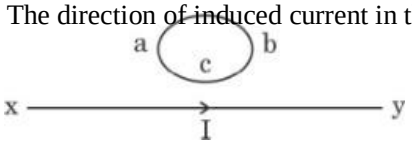
a) $[MT^{-2} A^{-1}]$

b) $[MLT^{-1}A]$

c) $[M^1L^2T^{-1}A^{-2}]$

d) $[M^2LT^{-2}A^{-1}]$

7. The direction of induced current in the loop abc is: **[1]**



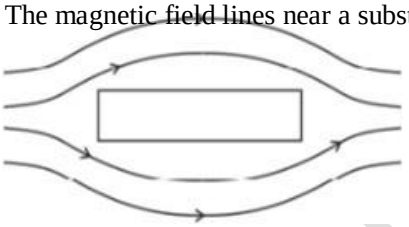
a) along abc if I increases

b) along acb if I increases

c) along abc if I is constant

d) along abc if I decreases

8. The magnetic field lines near a substance are as shown in the figure. The substance is: **[1]**



a) Copper

b) Sodium

c) Iron

d) Aluminium

9. The intensity ratio of the maxima and minima in an interference pattern produced by two coherent sources of light is 9 : 1. The intensities of the used light sources are in the ratio: **[1]**

a) 3 : 1

b) 9 : 1

c) 10 : 1

d) 4 : 1

10. Consider a neutral conducting sphere. A positive point charge is placed outside the sphere. Then the net charge on the sphere is - **[1]**

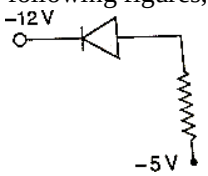
a) Negative and distributed non–uniformly over the entire surface of the sphere

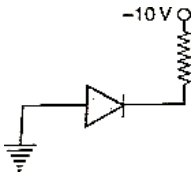
b) Negative and appears only at the point on the sphere closest to the point charge

c) Negative and distributed uniformly over the surface of the sphere

d) Zero

11. In the following figures, which one of the diodes is reverse biased? **[1]**

a) 

b) 

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12. Figure shows three transparent media of refractive indices μ_1, μ_2 and μ_3. A point object O is placed in the medium μ_2. If the entire medium on the right of the spherical surface has refractive index μ_1, the image forms at O'. If this entire medium has refractive index μ_3, the image forms at O''. In the situation shown in the figure given ahead: **[1]**

a) the image forms to the right of O''

b) the image forms to the left of O'

c) the image forms between O' and O''

d) two images form, one at O' and the other at O''

13. **Assertion (A):** The de Broglie equation has significance for any microscopic or sub-microscopic particle. **[1]**
Reason (R): The de Broglie wavelength is inversely proportional to the mass of the object if velocity is constant.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

14. **Assertion:** The capacity of a conductor, under given circumstances, remains constant irrespective of the charge present on it. **[1]**
Reason: Capacity depends on size and shape of a conductor and also on the surrounding medium.

a) Assertion and reason both are correct statements and reason is correct explanation for assertion.

b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.

c) Assertion is correct statement but reason is wrong statement.

d) Assertion is wrong statement but reason is correct statement.

15. **Assertion (A):** Thin films such as soap bubble or a thin layer of oil on water show beautiful colours when illuminated by monochromatic light. **[1]**
Reason (R): The colours are obtained by dispersion of light.

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a) Both A and R are true and R is the correct

b) Both A and R are true but R is not the correct

c) A is true but R is false

d) A is false but R is true.

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